Algebraic Geometry – Exercise Sheet 1

Week 1: Affine algebraic sets, projective space and localisation

Monday and Thursday afternoon sessions; Lectures 1–5

Difficulty. $\star$ routine; $\star\star$ synthesis; $\star\star\star$ challenge or extension.
Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

Monday afternoon — material available: Lecture 1 only

Exercise 1. Three points in the affine plane $\star$ CORE

Let

$$S = \{(0,0), (1,0), (0,1)\} \subset \Bbbk^2.$$

(i) Show that

$$\mathbf{I}(S) = (x, y) \cap (x-1, y) \cap (x, y-1).$$

(ii) Prove directly that

$$\mathbf{I}(S) = \big(xy,\, x(x-1),\, y(y-1)\big).$$

(iii) Show that evaluation at the three points induces an isomorphism

$$\Bbbk[x,y]/\mathbf{I}(S) \simeq \Bbbk^3.$$

Exercise 2. Two axes and their coordinate rings $\star$ CORE

In $\Bbbk^2$, set $X = \mathbf{V}(x)$ and $Y = \mathbf{V}(y)$.

(i) Show that

$$X \cup Y = \mathbf{V}(xy), \qquad X \cap Y = \mathbf{V}(x,y).$$

(ii) Compute the coordinate rings of X, Y, $X \cup Y$, and $X \cap Y$.

(iii) Explain geometrically why $\Bbbk[x,y]/(xy)$ has zero divisors.

Exercise 3. The parabola and the cusp $\star\star$ CORE

Consider

$$P = \mathbf{V}(y - x^2), \qquad C = \mathbf{V}(y^2 - x^3) \subset \Bbbk^2.$$

(i) Show that $t \mapsto (t, t^2)$ defines an isomorphism $\mathbb{A}^1 \to P$.

(ii) Show that $t \mapsto (t^2, t^3)$ defines a morphism $\nu : \mathbb{A}^1 \to C$ which is bijective on the underlying sets of $\Bbbk$-points.

(iii) Write the corresponding homomorphism of coordinate rings and prove that ν is not an isomorphism.

Exercise 4. Polynomial maps of the affine line $\star$ CHOICE

Show that morphisms of affine algebraic sets $\mathbb{A}^1_{\Bbbk} \to \mathbb{A}^1_{\Bbbk}$ are exactly polynomial maps $t \mapsto f(t)$. Describe the coordinate-ring homomorphism and check explicitly how composition reverses arrows.

Thursday afternoon — material available: Lectures 1–4

Exercise 5. Radicals and reduced rings SCAFARTO. $\star$ CORE

Compute

$$\sqrt{(x^2, y^3)}, \qquad \sqrt{(x^2 y^3)}$$

in $\Bbbk[x,y]$. Decide which of the rings

$$\Bbbk[x]/(x^2), \qquad \Bbbk[x,y]/(xy), \qquad \Bbbk[x,y]/(x^2, xy)$$

are reduced.

Exercise 6. Points at infinity $\star$ CORE

In the affine chart $x_0 \neq 0$ of $\mathbb{P}^2_{\Bbbk}$, write $x = x_1/x_0$ and $y = x_2/x_0$.

(i) Homogenise the parabola $y = x^2$ and determine its points at infinity.

(ii) Homogenise the hyperbola $xy = 1$ and determine its points at infinity.

(iii) Explain geometrically why the answers are different.

Exercise 7. The line through two projective points ⋆ CHOICE

Let $p = [a_0 : \cdots : a_n]$ and $q = [b_0 : \cdots : b_n]$ be distinct points of $\mathbb{P}^n_{\Bbbk}$. Show that the line through them is defined by the 3×3 minors of

$$\begin{pmatrix} a_0 & \cdots & a_n \\ b_0 & \cdots & b_n \\ x_0 & \cdots & x_n \end{pmatrix}.$$

Write the equation explicitly for $n = 2$.

Exercise 8. Points in $\mathbb{P}^2$ from minors ⋆⋆ CHOICE

Show that the 2×2 minors of

$$M = \begin{pmatrix} x & 0 \\ y & y \\ 0 & z \end{pmatrix}$$

define the three coordinate points in $\mathbb{P}^2$. For four lines $\ell_1, \dots, \ell_4$ with no three concurrent, show that the maximal minors of

$$\begin{pmatrix} \ell_1 & 0 & 0 \\ -\ell_2 & \ell_2 & 0 \\ 0 & -\ell_3 & \ell_3 \\ 0 & 0 & -\ell_4 \end{pmatrix}$$

define their six pairwise intersection points.

Exercise 9. Equations of the Segre and Veronese images ⋆⋆⋆ OPTIONAL

a) For

$$([s_0 : s_1], [t_0 : t_1]) \longmapsto [s_0 t_0 : s_0 t_1 : s_1 t_0 : s_1 t_1],$$

compute the kernel of $z_{ij} \mapsto s_i t_j$ and identify the image in $\mathbb{P}^3$.

b) For the quadratic Veronese map $\mathbb{P}^2 \to \mathbb{P}^5$, arrange the target coordinates as a symmetric 3×3 matrix and show that the image is defined by its 2×2 minors.

Exercise 10. Localising the coordinate axes ⋆ CORE

Let

$$A = \Bbbk[x, y]/(xy).$$

(i) Compute A_x and A_y.

(ii) Compute $A_{(x)}$ and $A_{(y)}$.

(iii) Explain what these localisations retain and what they discard geometrically.

Exercise 11. Localisation with a nilpotent ⋆ CHOICE

Let $A = \Bbbk[\varepsilon]/(\varepsilon^2)$. Compute

$$A_\varepsilon, \qquad A_{(\varepsilon)}, \qquad A_{1+\varepsilon}.$$

Explain why the three answers are so different.

Exercise 12. Two finite module structures ⋆⋆ CHOICE

Let

$$B = \Bbbk[x, y]/(y^2 - x).$$

Show that B is free of rank 2 over $\Bbbk[x]$ and free of rank 1 over $\Bbbk[y]$. Compare geometrically the corresponding morphisms to the affine line.

Exercise 13. Noether normalisation for the hyperbola ⋆⋆⋆ CHOICE

Let

$$B = \Bbbk[x, y]/(xy - 1), \qquad u = x + y \in B.$$

Show that B is finite over $\mathbb{k}[u]$. Deduce that the corresponding morphism to $\mathbb{A}^1_{\mathbb{k}}$ has fibres with at most two points.

After Friday's lecture or homework

Exercise 14. A presheaf which is not a sheaf ⋆ OPTIONAL

Let $X = U \amalg V$ be the disjoint union of two non-empty open subsets. Define a presheaf $\mathcal{F}$ by

$$\mathcal{F}(W) = \mathbb{Z} \quad \text{for } W \neq \varnothing, \qquad \mathcal{F}(\varnothing) = 0,$$

with the evident restriction maps. Show that $\mathcal{F}$ is not a sheaf.

Exercise 15. Continuous and holomorphic functions ⋆⋆ CHOICE

Let $U \subset \mathbb{C}$ be connected and contain at least two points. Show that $\mathcal{O}(U)$ has no zero divisors, whereas $\mathcal{C}^0(U)$ has non-zero functions f, g with $fg = 0$.

Exercise 16. $\mathcal{O}(d)$ on the projective line ⋆⋆⋆ OPTIONAL

Identify $\mathbb{P}^1_{\mathbb{C}}$ with $\mathbb{C} \cup \{\infty\}$. Define $\mathcal{O}(d)$ as the sheaf of meromorphic functions which are holomorphic away from ∞ and have at most a pole of order d there when $d \geq 0$; for $d < 0$, require a zero of order at least $-d$ at ∞. Compute its global sections.